



**THE UNIVERSITY
OF QUEENSLAND**
A U S T R A L I A

Fear of temperature

Computational analysis of policy, media
and public climate emotions

Project proposal

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Use of AI Statement

Topic exploration: Yes; AI supported discussion of the topic and candidate approaches.

Research questions: Yes; AI assisted formulation and critique of Section 3.1; I selected the research focus.

Literature review: Yes; AI assisted searches and selected claim/reference checks for Section 2. Reading depth is recorded separately.

Formatting and language: Yes; AI assisted English drafting, translation, editing and document layout.

Other uses: Methodological discussion, planning, Figures 1-8 and export code. No empirical research-data analysis is reported here.

I acknowledge the use of generative AI tools in completing this assessment. Details of which tools were used and how they were used are provided in the table below, along with appropriate in-text and full references. I take responsibility for critically evaluating and integrating the AI-generated content, and ensuring it adheres to academic integrity standards.

Table A. AI tool use.

Tool and recorded use	Main activities
ChatGPT; 11 and 15 September 2026	Topic and question development, literature assistance, methodological discussion, writing and language editing.
Gemini; date/version unrecorded	Suggested NLP modules and event/lag analysis; suggestions were assessed and revised by the author.
Codex; 15 September 2026	Source and feasibility checks, discussion records, drafting, figures, export code and layout inspection.

Tools: ChatGPT ([OpenAI, n.d.-a](#)), Gemini ([Google, n.d.](#)), Codex ([OpenAI, n.d.-b](#)). Model versions are retained in the working record where available. This declaration concerns proposal preparation; it reports no empirical research-data analysis.

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Figures: 1. Research relationships; 2. Competing mechanisms; 3. RQ evidence routes; 4. Source denominators; 5. NLP pipeline; 6. Measurement and timing; 7. Contextual evidence; 8. Calendar and workload.

Tables: A. AI use; 1. Literature synthesis; 2. Source register; 3a-b. Measures and annotation; 4. Event plan; 5. Evaluation; 6. Risk assessment; 7. OHS and ethics.

Reading note. This proposal describes planned research; it does not report completed experiments. Numerical examples and diagram trajectories explain the design. Access-dependent choices will be resolved through the documented pilot. The provisional submission date is 16 September 2026; the course deadline is 17 September 2026 at 15:00.

1. Introduction

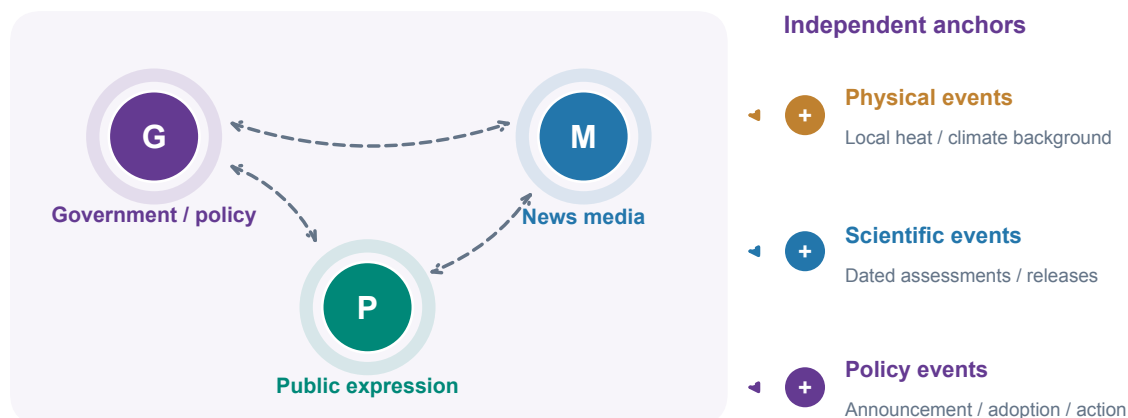
1.1 Motivation and significance

Rising temperature is both a physical process and an object of social anticipation. People may describe immediate danger during a heatwave, worry about the world their children will inherit, or assign responsibility for future disruption. These expressions involve different emotions, time horizons and social relationships. Climate-anxiety research recognises psychological responses to anticipated environmental change as well as direct exposure (Clayton, 2020). Studying how such responses are communicated can illuminate the changing social meaning of warming without assuming that every negative statement expresses fear.

Fear of Temperature asks how warming-related fear and anxiety develop across government and policy discourse, news media and public expression. A central problem is temporal ordering: does public concern follow physical events, institutional communication or media attention, and when might it precede them? Earlier work links climate concern to political cues, media and other conditions (Brulle et al., 2012). Research on affective agenda dynamics also demonstrates the relevance of comparing government, media and public expressions (Zhou et al., 2023). Figure 1 frames these as competing relationships to investigate rather than a predetermined chain of influence.

One social question, three discourse roles

Who changes first, and how is warming-related fear explained?



Dashed arrows: competing relationships, not established effects.

Figure 1. Three discourse roles. Dashed links are candidate relationships, not causal findings.

1.2 A methodological transition

This computational social science study investigates climate communication through a reproducible corpus, an NLP pipeline and temporal analysis. Earlier dictionary and n-gram exploration develops into natural-language collection, database construction and Transformer representations. We compare computational measurements against explicit baselines and contextual evidence, reducing discretionary passage selection while testing construct validity. The purpose remains the author's investigation of social emotions at scale; the principal scholarly outcome is the thesis.

1.3 Scope and historical framing

The core collection will prioritise English-language texts associated with the United States, European settings, Australia and New Zealand. Broad geographical coverage is an objective, while the main comparison concerns source roles and identifiable communities.

Chinese-language material, East Asian settings and AOSIS members are extension candidates rather than dependencies of the core study. Publication location, author location and the place discussed in a text will be distinguished. English language alone will not establish geographic or cultural identity.

The target collection begins in January 1988 and extends to a documented cutoff in 2026. The establishment of the IPCC in 1988 motivates an institutional anchor ([Intergovernmental Panel on Climate Change, n.d.](#)). It does not establish an emotional turning point. Recoverable material from 1938 onwards may form an earlier contextual layer, with Callendar's work providing a scientific-historical reference ([Callendar, 1938](#)). A claim about change across 1988 requires adequate observations on both sides; the main collection alone cannot demonstrate that transition.

Historical span and analytical coverage are therefore different. Each comparison will use a sufficiently dense common window, with incomplete months, archive gaps and changes of source explicitly recorded. Letters to editors and early forums may extend the record of public expression, but their selection processes differ from contemporary platforms. The intended inference concerns the sampled discourse. Claims about population emotion require additional sampling or validation evidence.

Longer-term anticipated consequences are the substantive priority. Direct heat danger will be collected alongside them and labelled after collection. Emotional category, immediate bodily-threat cues, time horizon and threatened group will be separate, potentially overlapping attributes. Fear can concern the future, and collective anxiety can arise during a present emergency. Cultural heritage may inform discussion, but the project does not presuppose heritage status.

1.4 Broad aim and intended contribution

The broad aim is to understand how warming-related social emotions, anticipated futures and responsibility narratives change across policy, news and public discourse, and how their temporal relationships vary with physical and institutional events.

Technical resource. A versioned, traceable corpus and reproducible pipeline with documented coverage, denominators, sampling weights and a data dictionary. Code, configurations and random seeds will support reproduction; only permitted material will be released.

Validated method. A modular workflow connecting relevance detection, emotion, relations and topics to temporal analysis, evaluated against baselines and across sources and periods. Any algorithmic novelty claim depends on demonstrated improvement, not simply combining existing models.

Empirical contribution. A thesis explaining temporal relationships and responsibility narratives through statistical estimates and contextual evidence, with explicit causal limits and potential for publication. Small or uncertain associations remain legitimate findings; success does not require a dominant driver or significant policy effect.

2. Background and Literature Review

2.1 Climate emotions and anticipated futures

Climate anxiety concerns responses to possible as well as experienced environmental harm. Clayton cautions against assuming that concern is necessarily maladaptive (Clayton, 2020). Solastalgia instead describes distress associated with environmental change in a place to which people remain attached (Albrecht et al., 2007). These accounts connect emotion to anticipated futures and lived environments, but neither makes all negative climate language equivalent to anxiety.

Pihkala's taxonomy distinguishes fear, worry, grief, anger, guilt and hope (Pihkala, 2022). For this project, that diversity argues against a single scale running from bodily panic to existential anxiety. Emotional category, target, time horizon and affected group need separate labels. Textual worry does not establish a persistent psychological condition; keyword cues require contextual validation.

O'Neill and Nicholson-Cole examine fear-inducing climate imagery and engagement (O'Neill & Nicholson-Cole, 2009). Their distinction between attracting attention and supporting engagement matters here: a rise in threatening representations need not indicate stronger policy support. The proposed measurements therefore separate issue attention, emotional expression and stance. This gives a more specific account of discourse than a negative-sentiment score, while retaining the limitation that expressions are not direct observations of population mental health.

2.2 Agenda-setting and responsibility frames

Agenda-setting addresses relationships between the prominence of issues in media and public agendas (McCombs & Shaw, 1972). A corpus share, however, measures the prominence of an issue in sampled output; without audience information it does not measure exposure or establish whose agenda changed.

Framing concerns how communication selects a problem, causal interpretation, evaluation and response (Entman, 1993). Increased climate coverage can frame warming as economic disruption, physical danger or intergenerational injustice. These are different interpretations even if their overall issue attention is similar. Cause, blame and response duty also differ: an actor may be asked to respond without being blamed for creating the problem. Relations and contextual evidence are therefore needed alongside frequency and emotion measures.

Similar policy and public language may reflect adoption, quotation or opposition; convergence alone does not establish agreement.

Table 1. Literature and design.

Evidence strand	What it contributes	Requirement for this study
Climate emotions	Multiple responses to threat	Separate emotion, target and horizon
Agenda and framing	Prominence versus interpretation	Measure attention and narrative content
Climate concern	Elite/media and physical explanations	Compare roles and exposure scales
Contextual NLP	Retrieval and structured text labels	Validate by source and period
Temporal methods	Ordering and event-related changes	State estimands and causal limits

2.3 Evidence from climate concern and role comparisons

Brulle and colleagues construct quarterly U.S. climate-concern measures from 74 surveys during 2002–2010, comparing weather, science information, media and political factors (Brulle et al., 2012). Their findings motivate institutional explanations, but aggregate concern is not specifically future-oriented fear. National quarterly observations may also conceal responses to local, short-lived heat exposure. The proposed text measures add narrative detail and shorter eligible windows; they introduce selection and classification errors that survey-based concern does not share in the same form. They should complement this evidence rather than be assumed more accurate.

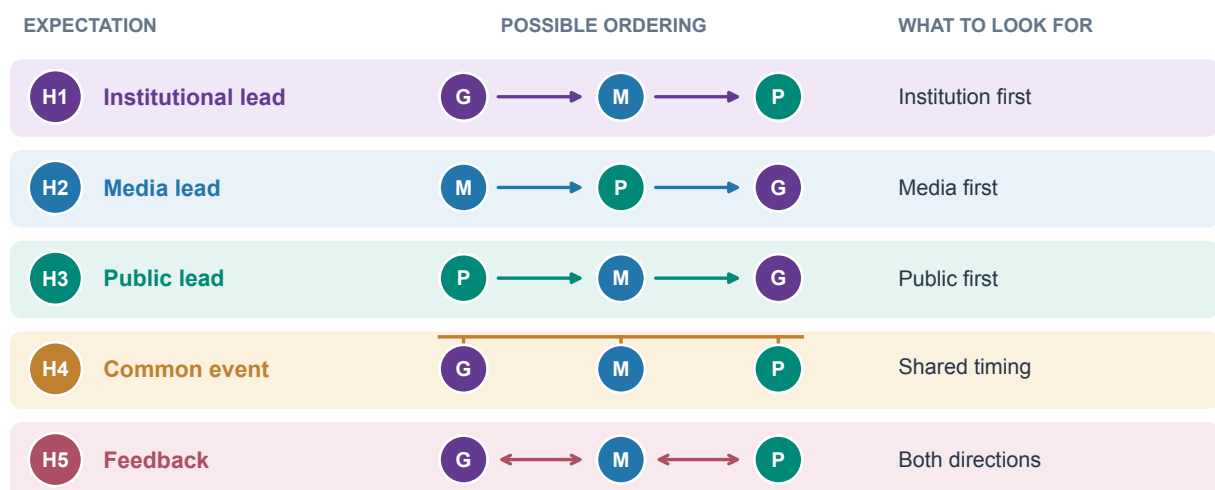
Zhou and colleagues compare public, government and media emotional communication on Weibo during COVID-19 (Zhou et al., 2023). This is a close precedent for role-based temporal comparison, but one platform and a pandemic period differ from a historical climate corpus spanning institutions and media systems. The present design retains the role comparison while testing source continuity, emotional targets and independent physical observations. Neither precedent establishes that a fixed direction of influence applies across all periods.

2.4 Competing temporal mechanisms and expectations

Institutional signals can supply authoritative risk interpretations (Brulle et al., 2012); media can select and frame issues (Entman, 1993; McCombs & Shaw, 1972); public expression may supply news material and pressure for response. Figure 2 makes the competing temporal expectations visible. H3 is a project hypothesis informed by role-comparison research (Zhou et al., 2023), not an established climate result.

Five competing temporal patterns

Read each row as a candidate explanation; routes may differ across windows.



G: government/policy M: media P: public · Predictive ordering alone is not causation.

Figure 2. Competing temporal mechanisms. Arrows indicate candidate orderings; colours identify roles.

The same ordering can arise from anticipation, shared releases, reposting or changing participants. A coarse monthly bin can hide a faster ordering. Local heat exposure also differs from global climate background. These expectations may coexist across windows; temporal prediction alone cannot identify social feedback or individual emotional contagion.

2.5 NLP for emotion, relations and temporal text analysis

Sentence-BERT supports efficient semantic comparison of sentences (Reimers & Gurevych, 2019). This can locate relevant expressions beyond exact dictionary phrases, but similarity does not by itself measure emotion, intensity or agreement. A shared encoder will maintain a common representation; validated relevance decisions will determine which texts enter the climate subset. A lexical/TF-IDF baseline remains useful for assessing what contextual representations recover and which relevant expressions they miss.

Aspect-based sentiment analysis links evaluations to their objects (Pontiki et al., 2014). This matters when a negative statement concerns the cost of a policy rather than warming. GoEmotions distinguishes fine-grained emotions in English Reddit comments (Demszky et al., 2020); it includes fear and nervousness, but no dedicated climate-anxiety category. Neither general sentiment nor an unvalidated model label can supply the project's future-worry measure. Cross-period and cross-source validation will assess this transfer before monthly means are interpreted.

PropBank describes predicate-argument roles (Palmer et al., 2005), while emotion-cause pair extraction links emotional expressions with reported causes (Xia & Ding, 2019). These inform complementary parts of the relation layer. A grammatical subject is not automatically culpable: in "government blamed industry", the speaker and blamed actor differ. The project must also distinguish described causes from attributed blame and response duties, retaining the predicate and evidence span rather than reducing every relation to an unqualified triple.

BERTopic combines embeddings, clustering and class-based term representation (Grootendorst, 2022). It provides a practical topic-analysis candidate, although the cited account is a preprint and suitability for this corpus remains to be tested. A change in topic prevalence can arise from source composition or vocabulary rather than changing meaning. A shared encoder, fixed-source checks, stable clusters and contextual passages are needed before describing semantic or metaphorical evolution.

The three modules jointly connect response and target (emotion), explanation (relations), and recurring narratives (topics). Their combination serves the substantive questions. Agreement is not independent confirmation when modules inherit the same retrieval or representation error.

2.6 Selected research gap

The research gap is a missing connection between role-specific temporal change, the emotion expressed and its attributed cause or responsibility. Attention studies do not by themselves establish whether fear increased, while emotion labels alone do not explain how a threat was framed. This project addresses the gap by joining validated attention, future-oriented emotion and responsibility measures to traceable policy, media and public passages, then comparing their timing within verified common coverage windows. The contribution is an integrated, auditable account of this relationship, not a claim that no previous study combined NLP and climate discourse.

The temporal methods answer bounded questions. Granger analysis concerns added prediction (Shojaie & Fox, 2022); ITS separates level and slope changes (Lopez Bernal et al., 2017), while a temporal cutoff alone does not identify a causal effect (Hausman & Rapson, 2018). Cross-lagged interpretations also require distinguishing stable unit differences from within-unit change (Hamaker et al., 2015). These limitations support the existing small-model pathway, rather than additional methods. The scholarly contribution depends on interpretable evidence and validated measurement, not a predetermined direction of influence.

3. Project Plan

3.1 Research questions

RQ1 - Who changes first? Which roles lead, lag or change together in warming-related attention and emotion? CCF and conditional VAR assess timing and held-out predictive gain over an own-history baseline, accounting for trends, seasonality and serial dependence.

RQ2 - What changes around events? How do heat, scientific and policy events relate to attention, emotion and framing? Segmented regression/ITS estimates level and slope changes, with window sensitivity and placebo-date checks.

RQ3 - How is fear explained? How are causes, blame, duties and threatened futures adopted, contested or reinterpreted across roles? Evidence-linked relations and topics support interpretation; span precision/recall and source/period holdouts test extraction quality.

Future-oriented warming fear or worry in public texts remains the main substantive outcome. The computational tasks make each question testable.

Research questions → computational tasks → evaluation

Three questions share one corpus, representation and validation protocol.

QUESTION	COMPUTATIONAL TASK	ANALYSIS + EVALUATION
RQ1 Who changes first?	Relevance + emotion → S by role; public E	CCF / conditional VAR → Lags + held-out predictive gain
RQ2 What changes at events?	Event links + emotion → S / E; common windows	Segmented regression / ITS → Level / slope + window checks
RQ3 How is fear explained?	Relations + topics → Cause / blame / duty	Relations + topic interpretation → Span P/R + source/period checks

Shared safeguards: fixed units + weights · source/time validation · original context

Figure 3. Research questions and computational evidence. Temporal comparisons use common bins; RQ3 retains passage-level evidence.

S measures attention; E measures emotion within relevant discourse. $B = S \times E$ is derived, not independent evidence. Prediction does not establish causation.

3.2 Evidence-driven implementation

The common-window pilot tests ingestion, role/date assignment, all three NLP modules and aggregation. Select by source overlap and independent dates. Expansion depends on measured processing and annotation costs.

Freeze outcomes, weights, labels, lag ranges and event windows before final comparisons.

Figure 5 shows the shared pipeline; secondary measures support interpretation.

3.3 Corpus construction and sampling

Table 2 separates archive history from obtainable coverage. Sources remain candidates until access and a sample extraction are verified.

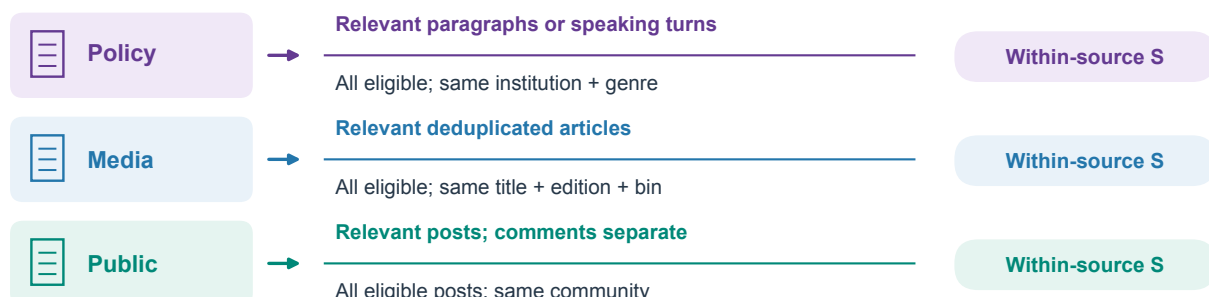
Table 2. Sources and coverage.

Candidate source	Role / type	Time coverage to establish	Access state
EPA, White House; UK/EU/AU/NZ records	Policy / institutional	Target 1988-2026; archive-specific gaps	● Candidate
UNFCCC; IPCC reports	Policy decisions; science	Publication-specific; discontinuous	● Candidate
NYT, Guardian, WSJ, FT; AU/NZ news	Media / articles	Title, edition and year entitlement	● Candidate
Letters, Usenet, preserved forums	Public / historical	Surviving dated records; no continuity assumed	● Candidate
Reddit research route	Public / platform	Rolling 5 years; 6-month delay (Reddit, n.d.)	● Candidate
Facebook, YouTube, Bluesky, Mastodon; forums	Public / platform	API/archive-specific; overlap unverified	● Candidate

Status: ● access/sample verified; ○ candidate; ○ unresolved. Record advertised, accessible and acquired dates, gaps and release conditions.

Three source roles, three auditable denominators

General attention comes from the eligible collection, not climate-only retrieval.



Climate-only subsets support emotion and relations; they cannot estimate general attention.

Figure 4. Attention denominators. Numerator and denominator share an eligible collection; posts and comments remain separate.

An issue-independent sample estimates attention. Enriched climate retrieval supports emotion/relations; prevalence requires known inclusion weights. Fixed stratum weights describe a stable source mix; size weights target the documented eligible population. Their effects will be checked.

Historical public channels remain separate unless overlap supports comparison. IPCC assessments retain a scientific subtype; climate-specialist collections cannot supply government-wide denominators. Missing bins differ from zero counts. Record IDs, publication/collection dates, parent documents, roles, quoted speakers, location evidence and duplicates to preserve provenance and control repeated-text weight.

3.4 Operational definitions and construct validity

The primary outcome is future-oriented warming fear or worry expressed in public texts. Fear, anxiety and worry will be treated as distinguishable, potentially overlapping expressions rather than diagnoses. Fear may concern an anticipated threat; worry often describes thinking about possible consequences. Neither category fixes a time horizon. Immediate heat danger and long-term concern will be collected together and labelled separately.

Table 3a. Constructs and measures.

Construct	Operational measure	Interpretation boundary
Attention S	Weighted relevant share of eligible units	Sampled output, not audience exposure
Emotion E	Validated label/share within relevant units	Expression, not clinical severity
Joint share B	Relevant emotional units / eligible units	Derived from S and E
Risk framing	Validated multi-label prototype/classifier scores	Mixed/other frames retained
Responsibility	Evidence-linked cause, blame and duty	Grammatical subject is not culpability
Stance	Position on a stated threat or policy	Policy opposition is not denial

For source stratum r and time bin t , U denotes the eligible unit set and i indexes its members. The weight w accounts for sampling; R indicates warming relevance and F with superscript future indicates expressed fear/worry targeting anticipated warming impacts. Equations (1)-(3) use identical units, weights and inclusion rules.

$$S_{rt} = \frac{\sum_{i \in U_{rt}} w_i R_i}{\sum_{i \in U_{rt}} w_i} \quad (1)$$

$$E_{rt} = \frac{\sum_{i \in U_{rt}} w_i R_i F_i^{\text{future}}}{\sum_{i \in U_{rt}} w_i R_i} \quad (2)$$

$$B_{rt} = \frac{\sum_{i \in U_{rt}} w_i R_i F_i^{\text{future}}}{\sum_{i \in U_{rt}} w_i} = S_{rt} E_{rt} \quad (3)$$

S separates discussion volume from E , the emotional composition of relevant discussion. B describes their joint prevalence and will not enter a model as independent evidence alongside both S and E . An empty denominator produces a missing value. Scores require calibration or an explicitly score-based interpretation; a classifier probability is not a validated measure of psychological intensity.

Weights and aggregation rules remain fixed within a comparison. Media article shares and policy paragraph shares can support within-series changes and temporal comparisons, but their levels are not identical quantities. Differences between such shares will not be interpreted as a direct ranking of which role is more fearful.

3.4 Operational definitions (continued)

Each annotated passage will retain its emotional target, temporal horizon, publisher/source role, quoted speaker, emotion-holder, and assertion/quotation scope. A newspaper can report a scientist describing residents' fear: these are three distinct positions. An institution's risk statement will not automatically count as an institutional emotion. Multiple or unknown holders remain explicit. Quoted public emotion stays attached to its institutional or media source; it is not silently transferred into the public-source series.

Negation and modality will be attached to the expression or relation they modify. A reassurance such as "should not panic" does not establish that panic occurred. A conditional future statement can express present worry about a possible event, so hypothetical wording is not a reason to discard the whole passage. Counterfactual and explicitly denied emotions require separate handling. Uncertain cases remain available for review and will not be forced into the positive class.

Table 3b. Annotation boundaries (synthetic examples).

Synthetic passage	Emotion / target / horizon	Speaker, holder and scope
I worry about my children's future.	Worry; target unspecified; future	Self-expression; climate context needed
I fear warming will harm my children.	Fear; warming impacts; future	Self-expression; anticipated harm
The government said people should not panic.	Reassurance; no observed fear established	Government reported; negated advice
Scientists warn heatwaves are deadly.	Risk statement; heat; unspecified time	Reported warning; no fear required
If warming worsens, I fear our town will become unsafe.	Fear; warming impacts; future	Present fear; conditional consequence
I am not afraid of warming; I oppose this tax.	Fear denied; policy opposition	Negation of fear; distinct stance

Time horizon and threatened group will be coded independently: immediate/near-term versus longer/intergenerational futures, and self/family/community where expressed. Ambiguous phrases will retain an unspecified category. The guide will record evidence spans and counterexamples for each label; development and held-out checks will be separated. This is a text annotation convention to be validated, not a new clinical taxonomy.

Semantic relevance will use several natural-language prototypes and development-set thresholds. Frame directions are not assumed orthogonal; the absolute projection onto one vector can assign the same magnitude to opposite directions and is not sufficient validation. Continuous alternatives face the same construct checks as classification. The fear/anxiety ratio and stance-distribution bimodality remain secondary: unstable denominators and classification mixtures can distort either, and bimodality alone does not demonstrate population polarisation.

These distinctions connect RQ3 back to the temporal results. A public-emotion rise will be examined alongside evidence about what was feared, who was described as affected, and whether the text assigned a cause, blame or duty. An unexpressed attribution will remain missing, rather than being inferred from a nearby event date.

3.5 Representation and three NLP components

We evaluate a modular NLP pipeline: relevance detection feeds three required components for relations, emotion and temporal topics. A shared sentence encoder produces versioned embeddings linked to context. We compare lexical/TF-IDF retrieval with Sentence-BERT using precision, recall, F1 and missed cases (Reimers & Gurevych, 2019), then freeze the encoder within comparisons. We report throughput (documents/hour), peak memory, runtime and annotation cost. Selection also depends on licence and local compatibility; MPS acceleration remains conditional. Shared representation does not imply joint multi-task training.

A modular NLP pipeline with shared representations

Relevance: lexical/TF-IDF versus Sentence-BERT; freeze the encoder after validation.

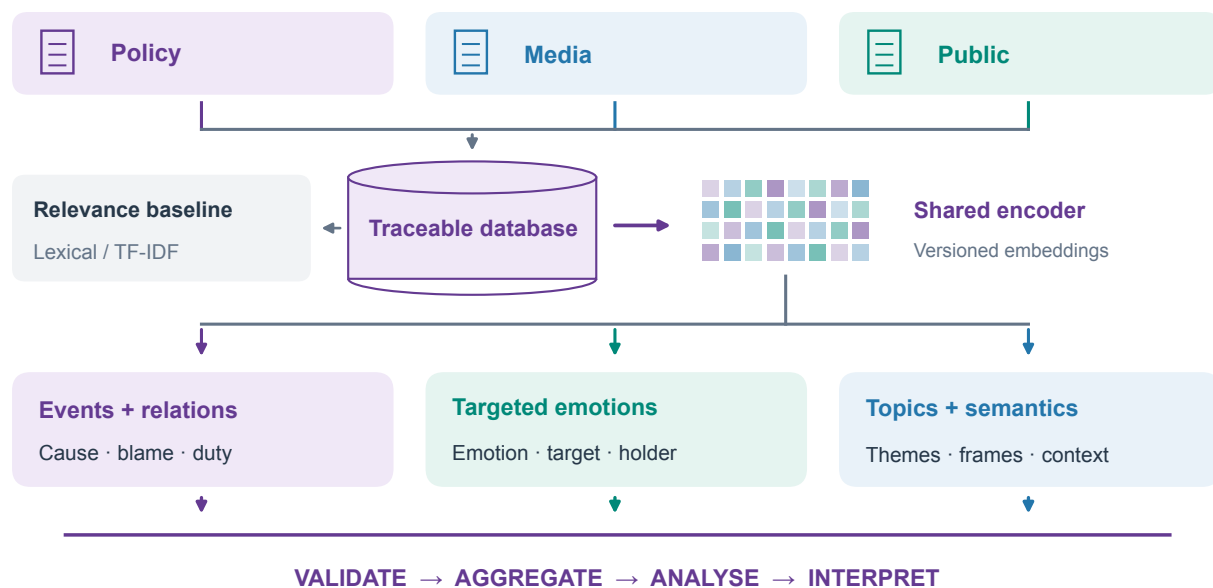


Figure 5. NLP pipeline. Context and provenance accompany all three components; validation precedes interpretation.

Event and relation extraction. Named entities and parsing support event candidates, followed by an evaluated semantic role labelling (SRL) or contextual relation model. Outputs distinguish event, speaker, described cause, blamed actor, response duty and affected group (Palmer et al., 2005; Xia & Ding, 2019). Evidence spans, negation, modality and quotation remain attached. Events used as external analytical dates are verified independently of narrative extraction.

Aspect-specific sentiment and emotion. A suitable GoEmotions-derived candidate will be compared with interpretable cues (Demszky et al., 2020). The aspect is the object being evaluated; the emotion layer distinguishes fear and project-defined future worry. Nervousness will not simply be renamed climate anxiety. The same procedure will check risks reported by institutions and emotions expressed by identifiable holders.

Temporal topics and semantics. BERTopic will use a shared embedding space and inspected representations (Grootendorst, 2022). Topic prevalence, frame distributions and representative passages will be compared by role and time, with resampling stability and outlier coverage checks. Matched-source context is required before interpreting a shift as changing meaning. The three components remain the agreed method set; validation determines their usable distinctions.

3.6 Temporal alignment and analysis

Comparisons require common coverage and frequency: monthly where supported, daily/weekly for dense heat pilots, or common quarters for sparse institutions. Recompute ratios from aggregated counts; never repeat quarterly values into artificial months. Regional GISTEMP anomalies describe climate background; station/ERA5 measures address local exposure (European Centre for Medium-Range Weather Forecasts, n.d.; NASA Goddard Institute for Space Studies, n.d.). Record location evidence, baselines and versions; publication location is not exposure.

RQ1: ordering and conditional prediction. Cross-correlation (CCF) explores candidate lags after addressing trends, seasonality and serial dependence. In Equation (4), X and Y are processed series; positive k means X precedes Y. Compare role attention, or policy/media attention with public emotion.

$$C_{XY}(k) = \text{Corr}(X_t, Y_{t+k}) \quad (4)$$

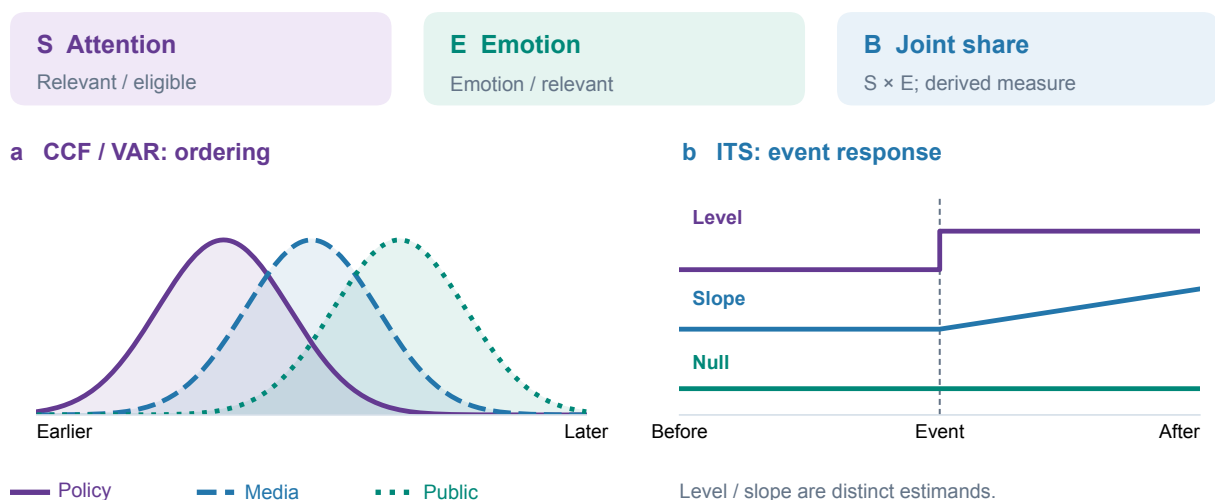
A small vector autoregression (VAR), where observations and diagnostics permit, will test whether other roles add predictive information beyond the outcome's own history (Shojaie & Fox, 2022). Report direction, lag range, uncertainty and held-out predictive gain. The largest CCF peak alone will not choose the lag order. Separate role-specific S from public E; B is secondary. Freeze lag limits, outcomes and processing before final comparisons.

RQ2: event-associated change. Segmented regression/interrupted time series (ITS) estimates a level change and a post-event slope change (Lopez Bernal et al., 2017). Equation (5) uses outcome Y and time t centred on event onset. D switches from 0 to 1 at onset; β_2 is the level change and β_3 the slope change. Disturbance u requires serial-dependence diagnostics and appropriate uncertainty treatment.

$$Y_t = \beta_0 + \beta_1 t + \beta_2 D_t + \beta_3 (tD_t) + u_t \quad (5)$$

Measure the discourse, then test its timing

Illustrative trajectories only; no observed data or estimated effects.



Inputs: aligned S / E series and event dates. Outputs: lags, predictive gain and level / slope.

Figure 6. Temporal analysis. All curves are schematic, not observed results.

3.6 Event selection and narrative interpretation (continued)

Select events from independent records before inspecting outcome peaks. Require a verified date, common coverage and an auditable denominator; pilot one eligible event. Distinguish announcement, adoption, implementation and scientific release dates.

Table 4. Candidate events.

Candidate / type	External date	Window / outcome	Comparison and limit
Kyoto adoption / policy	11 Dec 1997	Initially ±6 months; S/E level and slope	Historical overlap unverified
Paris adoption / policy	12 Dec 2015	Initially ±6 months; S/E level and slope	Anticipation; concurrent coverage
Heat episode / physical	○ Independent onset, peak and end	Dense local window; danger and future worry	Matched exposure; seasonality
IPCC release / scientific	○ Report-specific release date	Common coverage; S/E and frames	Shared shock; media spillover

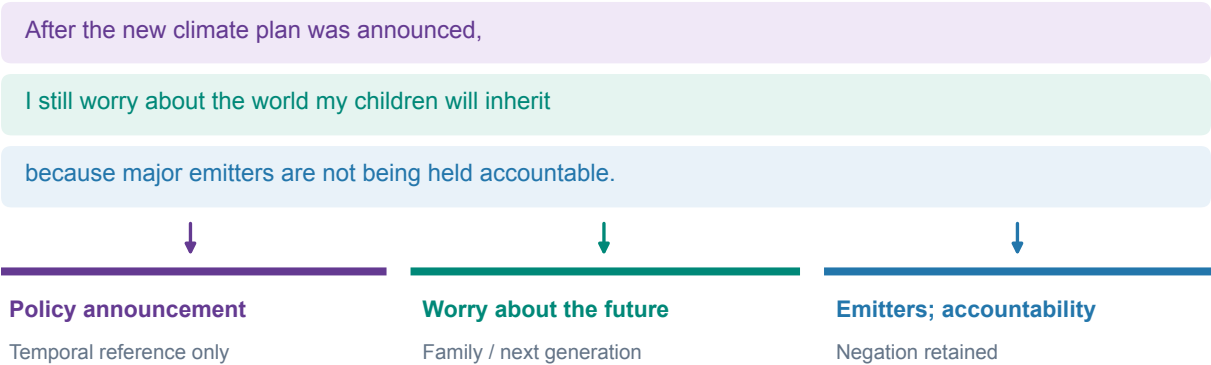
The adoption dates follow UNFCCC records (United Nations Framework Convention on Climate Change, n.d.-a, n.d.-b). Six-month windows are initial specifications, not guaranteed adequate time series. Longer baselines are needed where trend, seasonality or serial dependence cannot be assessed within them. Inferential windows and aggregation will be fixed before final tests, using coverage and seasonal-baseline adequacy rather than outcome peaks.

Comparison series require comparable pre-event measurement and defensibly different exposure. Another outlet/topic is not automatically a control. Register concurrent shocks and anticipation; inseparable events form a joint window. Placebo dates and alternative windows test sensitivity, not causal identification (Hausman & Rapson, 2018).

RQ3: narrative explanation. Relations and topics locate causes, blame, duties and threatened futures. Check quotation and stance before interpreting adoption, contestation or reinterpretation. Narrative attribution differs from estimated event effects; retain representative and contradictory passages.

One passage, several linked pieces of evidence

Synthetic example; contextual relations are not verified causal effects.



Still unresolved: policy caused the worry; who has the duty to act.
Figure 7. Evidence-linked annotation. Synthetic text illustrates labels and attribution; timing does not establish causation.

3.7 Evaluation and validity

A development batch of approximately 150 passages, balanced across roles, will expose label ambiguities and estimate annotation cost. It is not a powered final test. We size a separate stratified test set using class prevalence, annotation cost and desired precision. Enriched samples for rare labels will document their sampling implications.

The author will prepare an annotation guide and seek an independent second reader for a subset. Agreement will be reported only where independent ratings exist. If a second reader is unavailable, a separated repeat assessment and documented disagreements will be reported as limited single-author validation, without claiming inter-rater reliability. Development and test sets will be separated by document/duplicate cluster; source and period holdouts will probe transfer.

Table 5. Computational evaluation plan.

Task	Baseline / reference	Candidate / comparison	Metric / output	Validation setting
Relevance	Lexical / TF-IDF	Sentence-BERT retrieval	Precision, recall, F1	Source/period holdout; background misses
Emotion	Interpretable cues	GoEmotions-derived candidate	Macro/per-label F1; calibration	Annotated targets/holders; source/period transfer
Relations	Predicate/entity rules	SRL + contextual relations	Span and relation P/R	Annotated evidence; source/period transfer
Topics	Matched-source reference	BERTopic configurations	Stability, coverage; coherence review	Resampled texts; contextual passages
RQ1	Own-history model	Conditional VAR + roles	Held-out MSE gain; lag uncertainty	Common window; chronological holdout
RQ2	Pre-event trend	Segmented regression / ITS	Level/slope; uncertainty	Event windows; placebo dates
RQ3	Annotation guide	Relations + topic evidence	Agreement; traceable spans	Quotation/stance checks; counterexamples

Comparisons use identical eligible test texts and documented configurations. P/R denotes precision/recall; MSE is mean squared error. Corpus audits verify dates, roles, duplicate clusters and usable coverage. Topic and RQ3 interpretations require contextual evidence.

F1 below 0.70 or two-rater kappa below 0.65 may trigger review during development; these are provisional warning levels, not universal adequacy standards. Passing them alone will not validate the aggregate measures. Classification error will also be tested for its impact on temporal estimates through resampling or plausible error perturbations. Time-block and source/author/duplicate dependence will inform uncertainty estimation.

The primary analysis plan will freeze outcomes, thresholds, weighting and lag ranges before final comparisons. Alternative windows and models will be identified as sensitivity or exploratory analyses. Non-significance will not be interpreted as proof of no physical influence or exclusive policy control (Wasserstein & Lazar, 2016).

3.8 Project risks and scope adjustment

Database construction is the main delivery risk: delays consume validation and analysis time. Table 6 links observable triggers to specific fallbacks. Ratings indicate management priority, not estimated probabilities; Section 3.10 covers ethics and OHS.

3.8.1 Risk assessment

Table 6. Risks, triggers and fallbacks.

Risk	Trigger / consequence	Rating	Concrete fallback
R1. Database overrun	Core freeze missed: analysis time shrinks.	High	At 18 December, stop optional ingestion; freeze validated batches and reduce source/window breadth.
R2. Access or release	Access denied, delayed or incompatible with reuse.	High	If Reddit is unavailable at the pilot gate, assess a permitted Mastodon community; restrict comparison to its verified common window.
R3. Sparse coverage	Low effective counts or insufficient common bins.	High	Use common non-overlapping quarters; otherwise shorten the window. Sparse historical texts remain contextual evidence.
R4. NLP validity	Source/period audit misses prespecified quality targets.	High	Refine labels and revalidate; narrow unreliable distinctions. Report unsupported constructs qualitatively rather than substitute unvalidated scores.
R5. Temporal validity	Diagnostics fail or events overlap; estimates unstable.	High	Reduce lag/model complexity or retain descriptive CCF/event profiles with uncertainty. Null results do not establish another driver.
R6. Composition drift	Source mix or encoder changes produce a break.	High	Recompute on a fixed-source subset and frozen encoder; separate incompatible periods rather than splice trends.
R7. Compute / review	Pilot throughput or reviewer availability misses budget.	Medium	Cache batches, cap comparisons and use CPU inference; reduce validation sample breadth and disclose any single-reader limitation.
R8. Disruption / loss	Illness, corruption or failed restore threatens delivery.	Medium	Restore tested, versioned backups; prioritise core analyses with the supervisor and use protected contingency.

3.8.2 Review points and scope adjustment

Mastodon is a candidate fallback, not a guaranteed archive: instance permissions, dated coverage and an all-topic denominator must be verified ([Mastodon, n.d.](#)). Permitted letters or forum archives may support earlier public strata. These sources will not be spliced into a continuous population series. If no public source qualifies, narrow the three-role comparison to a supported window; document any RQ left unanswered.

Review gates are 16 October (pilot), 18 December (core corpus), 15 January (validation) and 28 February 2027 (all analyses). A missed gate or sustained workload above 10-20 hours/week triggers scope review. Protect the three roles, three NLP components and validation. Counts below 100 flag review; effective counts, uncertainty and usable time points determine aggregation.

3.9 Milestones, resources and time allocation

At 10-20 hours/week, the provisional budget is 450-590 hours plus 15% contingency. All analysis and robustness checks must finish by 28 February 2027. The 320-400 hours allocated before March require about 14-17 hours/week before contingency; sustained lower capacity will trigger early scope reduction.

A research calendar with protected analysis time

10–20 hours/week · planned task effort 450–590 h + 15% contingency

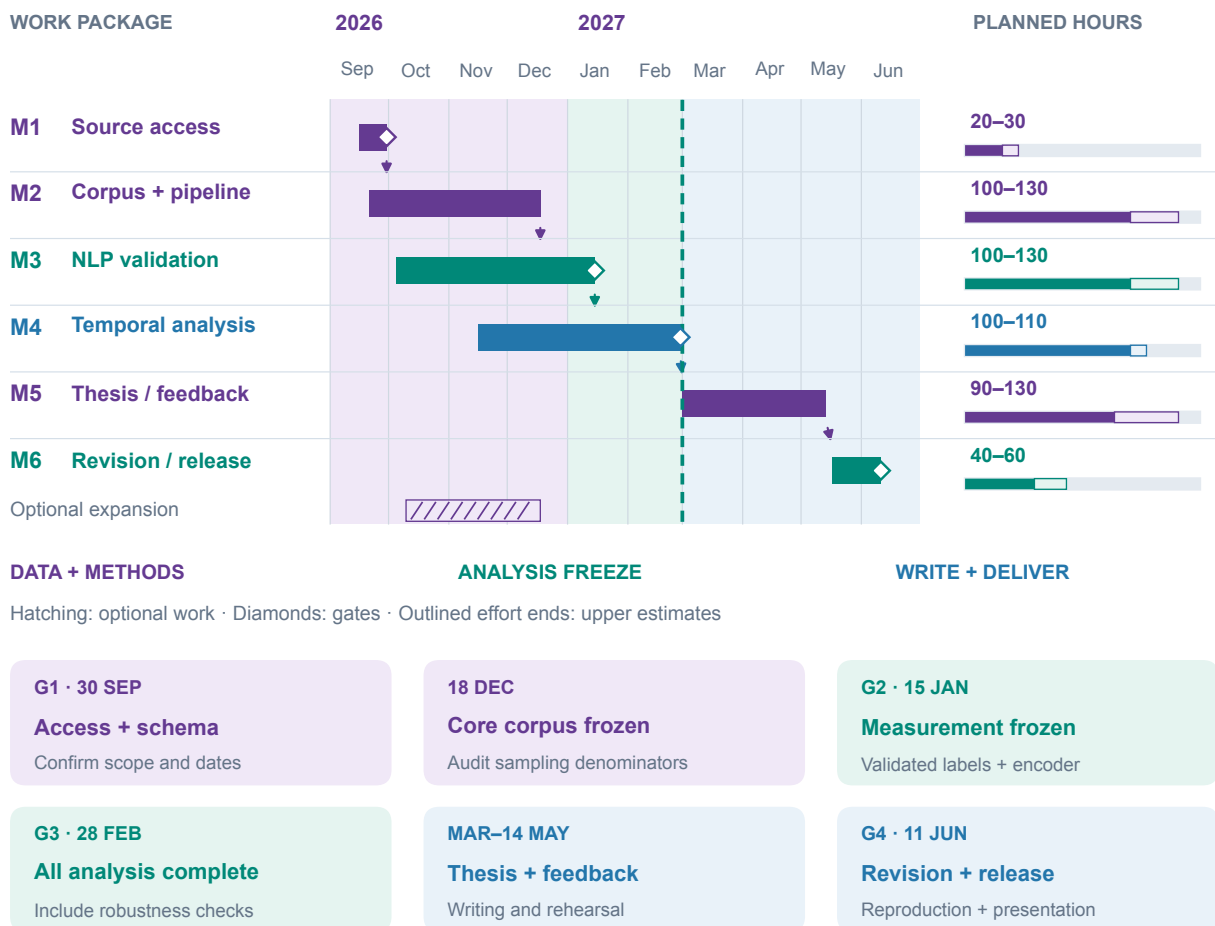


Figure 8. Work programme and effort. Solid/outlined bars show lower/upper estimates; hatching denotes optional work. Hours are planned.

Resource dependencies are permitted archives, storage/local compute, annotation time and supervisory feedback. The pilot is due by 16 October; core data and denominator checks freeze on 18 December. Measurement validation freezes on 15 January, followed by all temporal, event, topic and attribution analyses by 28 February. Release requires a permissions review and reproduction instructions.

Assessment anchors: seminar, 12-16 October; Thesis Plan, 1-19 March 2027; rehearsal, 10-14 May; final pitch/poster/demonstration/Q&A, 7-11 June ([The University of Queensland \[UQ\], 2026](#)). March-June centres on writing and feedback using completed analyses.

Missed freezes trigger narrower comparisons while preserving three roles, three modules and the February target.

3.10 OHS, ethics and data governance

This computer-based documentary study involves no planned fieldwork or participant recruitment. Workstation, data-handling and ethics requirements will be reviewed with the supervisor before relevant work.

Table 7. OHS and data controls.

Issue	Preventive control	Review / residual issue
Posture and repetitive work	Ergonomic setup, breaks and task rotation	Adjust if discomfort develops.
Workload / distressing text	Bounded sessions and annotation batches; university support	Monitor fatigue and distress.
Personal data / quotations	Minimise identifiers; restrict raw access; assess quotation reidentification	Public access is not blanket consent.
Group representation	Use observable source roles; retain uncertainty	Do not infer sensitive identities or population prevalence.
Services / release	Review licences, transfer, retention and deletion; favour local processing	Features and embeddings may remain restricted.

Supervisory review. Before accessing human-authored datasets, Dai Pan will obtain Mashhuda Glencross's documented review of permissions, identifiability, storage and release. This project gate is separate from formal clearance. UQ students cannot be Chief Investigator (CI); the supervisor will submit as CI where appropriate (UQ, n.d.-c).

UQ determination. Seek advice from Research Ethics and Integrity (humanethics@research.uq.edu.au) on research involving people or their data, and document the outcome. Public availability does not establish exemption; intended publication cannot rely on the teaching-only exception. Apply through MyResearch for lower-risk or Human Research Ethics Committee (HREC) review, or register an eligible exemption. Relevant work begins only after clearance or exemption conditions are established (UQ, n.d.-a, n.d.-b). No approval is claimed here.

Later human evaluation. If human A/B evaluation is proposed, the supervisor will first review the protocol, recruitment, consent, tasks and data plan. Whether reviewers become research participants will be checked with Research Ethics and Integrity. Any required new approval or amendment must precede recruitment and data collection; a supervisor's quality check is not institutional approval (UQ, n.d.-a).

Restricted originals remain separate from permitted derivatives. Storage, external inference and release require source-specific checks of permissions, retention and deletion conditions.

3.11 Outputs, interpretation and research record

The database will record source-time keys, permitted IDs/URLs, weights, missingness, climate metadata, model versions, features and uncertainty. Permitted CSV/Parquet exports and code will include a data dictionary and versioned configuration; restricted data will have access descriptions.

The thesis will connect temporal findings to validated passages, including contradictory evidence and limits of population inference. A decision log, source register and progress notes will support supervisory discussion. Reproducible evidence will underpin subsequent publication or algorithmic claims.

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